Continuous Convergence and Curried Functional Spaces

1 Introduction

There are essentially two ways to represent a multivariable function. One is to complicate the domain, resulting in $f : A \times B \to C$, and the other is to complicate the codomain by writing $g : A \to C^B$, such that $g(a)$ is again a function that maps B to C. In programming, the latter approach is called *partial application* or *currying,* and we will adopt this terminology.
Classical multivariable calculus is built on the former, "uncurried" approach to defining functions of multiple arguments, and in this paper we seek a way to express calculus in the curried language instead.
We will be using the topological language of *filters* throughout the work, to simplify proofs. For the definition and properties of filters, see Appendix A.

2 General curried functions

Definition 2.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, ...\}$. The set $H_n(\mathbb{R})$ of *curried n-variable functions* is a topological vector space defined recursively as follows:

- If $n = 0$, we set $H_0(\mathbb{R}) = \mathbb{R}$;
- If $H_n(\mathbb{R})$ is defined, we let $H_{n+1}(\mathbb{R})$ to be the set of all functions from $\mathbb{R}$ to $H_n(\mathbb{R})$. This set is given the pointwise linear structure and the topology of pointwise convergence (see Appendix B), such that a filter $F \in \mathcal{F}(H_{n+1}(\mathbb{R}))$ converges to a function $f \in H_{n+1}(\mathbb{R})$ if and only if $F(x) \to f(x) \in H_n(\mathbb{R})$ for all $x \in \mathbb{R}$. Lemma 2.3 and Lemma 2.4 from Appendix B ensure that $H_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space.

Definition 2.2. Let $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ be a function. Define

$$\gamma\left(\tilde{f}\right)(x_1)(x_2)...(x_n) = \tilde{f}(x_1, x_2, ..., x_n).$$

Then $\gamma : \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}) \to H_n(\mathbb{R})$ is clearly a bijection. The function $f = \gamma\left(\tilde{f}\right)$ will be called the *curried version* of $\tilde{f}$.

Remark 2.3. Let $n \in \mathbb{N}$. If $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R})$ is given the topology of pointwise convergence together with the pointwise linear structure, then $\gamma : \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}) \to H_n(\mathbb{R})$ clearly becomes an isomorphism of topological vector spaces.

Lemma 2.4. *For every $n \in \mathbb{N}_0$, the space $H_n(\mathbb{R})$ is locally convex.*

> *Proof:* We employ induction over n. If $n = 0$, the statement is clear. Suppose that the space $H_{n-1}(\mathbb{R})$ is locally compact. It is then easy to see that $V(0, U)$ is a convex neighborhood of 0 in $H_n(\mathbb{R})$, where U is the convex neighborhood of $0 \in H_{n-1}(\mathbb{R})$ that exists by the induction hypothesis. ■

3 Continuous curried functions

Definition 3.1. Let $n \in \mathbb{N}_0$. The set $C_n(\mathbb{R})$ of *curried continuous functions* is a Hausdorff topological vector space defined recursively as follows:

- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$.

- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$, endowed with pointwise linear structure and the compact-open topology (see Appendix C of the Appendix). Lemma 3.5 and Lemma 3.6 ensure that $C_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space, provided that $C_n(\mathbb{R})$ is.

Lemma 3.2. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k\in\mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ if and only if $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x$ in X.*

Proof: First, let F be the derived filter of $\{f_k\}_{k\in\mathbb{N}}$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k\in\mathbb{N}}$ converging to $x \in X$, with its derived filter P. Now let V be a neighborhood of $f(x)$ in Y. By Proposition 3.3, we have $F(P) \to f(x)$, and so $V \in F(P)$. This means that V contains an element of the base of $F(P)$, namely a set

$$B_{n_0,m_0} = \{f_n(x_m) \,|\, n \geq n_0,\ m \geq m_0\}.$$

Hence, for $k \geq \max(n_0, m_0)$, we have $f_k(x_k) \in V$, as desired.

Conversely, suppose that $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x \in X$. First let us show that for any subsequence $\left\{f_{n_k}\right\}_{k\in\mathbb{N}}$, we have $f_{n_k}(x_k) \xrightarrow[k\to\infty]{} f(x)$.
To this end, let $n \in \mathbb{N}$. We let $z_n = x_{\varphi(n)}$, where $\varphi(n)$ is the minimal number of those $k \in \mathbb{N}$ that satisfy $n \leq n_k$. We clearly see that $n_1 \geq n_2$ implies $\varphi(n_1) \geq \varphi(n_2)$ and that $\varphi(n_k) = k$ for all $k \in \mathbb{N}$. Therefore, the sequence $\{z_n\}_{n\in\mathbb{N}}$ converges to x. Hence we have $f_k(z_k) \xrightarrow[k\to\infty]{} f(x)$, and so

$$f_{n_k}(x_k) = f_{n_k}\left(z_{n_k}\right) \xrightarrow[k\to\infty]{} f(x). \tag{1}$$

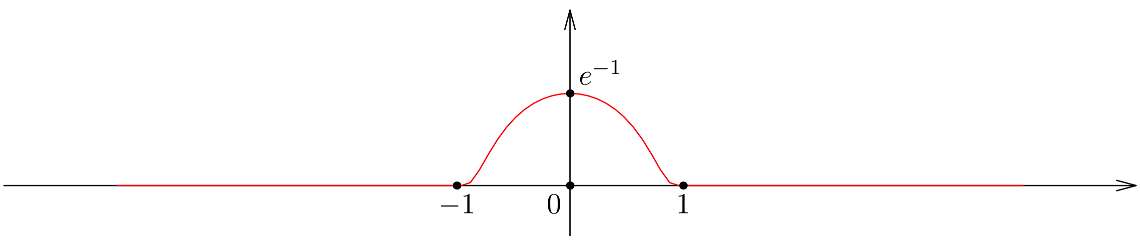
Now, to show that $f_k \to f$ in $\mathcal{C}(X, Y)$, we need to show that the derived filter F generated by the sets $C_n = \{f_k \mid k \geq n\}$ converges to f. To this end, let P be a filter in X converging to a point $x \in X$. Note that since X is first countable, the filter P has a countable base $\{A_m\}_{m\in\mathbb{N}}$ of neighborhoods of x. To show that $F(P)$ converges to $f(x)$, assume the contrary. Then some neighborhood V of $f(x)$ does not lie in $F(P)$, which is to say that for all $n, m \in \mathbb{N}$, we have $C_n(A_m) \not\subset V$. That means that for all $n, m \in \mathbb{N}$, there are $k_m \geq m$ and $x_m \in A_m$ such that $f_{k_n}(x_m) \notin V$.
In particular, we have $f_{k_n}(x_n) \notin V$ for all $n \in \mathbb{N}$. At the same time, since the sets A_n generate P, we see that $x_n \xrightarrow[n\to\infty]{} x$, so we must have $f_{k_n}(x_n) \xrightarrow[n\to\infty]{} f(x)$ by (1), a contradiction. ■

Corollary 3.3. *From the above lemma we conclude that if $n \geq 1$, a sequence of functions $\{f_k\}_{k=1}^\infty$ in $C_n(\mathbb{R})$ converges to $f \in C_n(\mathbb{R})$ if and only if for any sequence $\{x_k\}_{k=1}^\infty \subset \mathbb{R}$ converging to $x \in \mathbb{R}$, we have $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ in $C_{n-1}(\mathbb{R})$.*

Example 3.4. The topology on $C_n(\mathbb{R})$ is finer than the topology induced from $H_n(\mathbb{R})$, and does not coincide with it, unless $n = 0$. To see this, take $n = 1$ and consider the classical function

$$\varphi(x) = \begin{cases} \exp\left(\frac{1}{|x|^2-1}\right), & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

This is a $C^\infty(\mathbb{R})$ function supported on $[-1, 1]$. Its graph looks as follows:



Now consider the following sequence of functions:

$$f_k(x) = e \cdot \varphi\Big(k \cdot \Big(x - \frac{1}{k}\Big)\Big).$$

Clearly, f_k is infinitely differentiable on $\mathbb{R}$, supported on $[0, 2/k]$, and has a maximum at $x = 1/k$ with $f_k(1/k) = 1$. It is clear that for all $x \in \mathbb{R}$, we have $f_k(x) \xrightarrow[k\to\infty]{} 0$, so $f_k \to 0$ in the topology induced from $H_1(\mathbb{R})$. However, f_k do not converge to 0 in the native topology of $C_1(\mathbb{R})$. To see this, take $x_k = 1/k$ and observe that $f_k(x_k) \equiv 1 \neq 0$.

A similar argument applies for any other $n \geq 2$. Hence we conclude that $C_n(\mathbb{R})$ is not a subspace of $H_n(\mathbb{R})$.

Proposition 3.5. *Let $n \in \mathbb{N}_0$. Then the map*

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \to C_n(\mathbb{R})$$

defined by $\gamma(f)(x_1)(x_2)...(x_n) = f(x_1, ..., x_n)$, is well-defined and is a homeomorphism.

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$. By Proposition 3.4, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ■

Lemma 3.6. *For every $n \in \mathbb{N}_0$, the space $C_n(\mathbb{R})$ is locally convex.*

Proof: An easy proof by induction over n, similar to the case of $H_n(\mathbb{R})$. ■

Definition 3.7. A topological vector space is called a *Fréchet space* if its topology is compatible with a complete invariant metric.

Lemma 3.8. *For all $n \in \mathbb{N}_0$, the space $C_n(\mathbb{R})$ is a Fréchet space.*

Proof: If $n = 0$, the statement clearly holds. Now, assuming that $C_{n-1}(\mathbb{R})$ is a Fréchet space, we easily conclude by Lemma 3.8 that $C_n(\mathbb{R})$ is also a Fréchet space. ■

4 Differentiable functions

Definition 4.1. Let $m \in \mathbb{N}_0$ and $n \in \mathbb{N}_0$. The set $D_n^m(\mathbb{R})$ of *curried m times differentiable functions* is a topological space defined recursively as follows:

- If $n = 0$, we set $D_0^m(\mathbb{R}) = \mathbb{R}$ for all m, as usual;
- If $m = 0$, we set $D_n^0(\mathbb{R}) = C_n(\mathbb{R})$;
- Let $m, n > 0$, and assume that both $D_{n-1}^m(\mathbb{R})$ and $D_n^{m-1}(\mathbb{R})$ are defined. Then we let $D_n^m(\mathbb{R})$ be the set of all functions $f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ such that there is a function $f' \in D_n^{m-1}(\mathbb{R})$ which satisfies

$$\frac{f(x+h) - f(x)}{h} \xrightarrow[h\to 0]{} f'(x) \in H_{n-1}(\mathbb{R})$$

for all $x \in \mathbb{R}$, where **the convergence is taken in the space $H_{n-1}(\mathbb{R})$**. Clearly, if $f, g \in D_n^m(\mathbb{R})$ and $\alpha, \beta \in \mathbb{R}$, we have

$$(\alpha f + \beta g)' = \alpha f' + \beta g',$$

so in particular, $\alpha f + \beta g \in D_n^m(\mathbb{R})$.
We endow $D_n^m(\mathbb{R})$ with the coarsest topology in which the maps

$$\begin{aligned} i : D_n^m(\mathbb{R}) &\to \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})), & \quad d : D_n^m(\mathbb{R}) &\to D_n^{m-1}(\mathbb{R}) \\ f &\mapsto f & f &\mapsto f' \end{aligned} \tag{2}$$

are continuous. This topology will be called the *differential topology of order m*. We prove below that $D_n^m(\mathbb{R})$ are Hausdorff topological vector spaces.

Lemma 4.2. *For all $m,n \in \mathbb{N}_0$, we have $D_n^{m+1}(\mathbb{R}) \subset D_n^m$, and the inclusion map*

$$j : D_n^{m+1}(\mathbb{R}) \to D_n^m(\mathbb{R})$$

is continuous, meaning that the topology on $D_n^{m+1}(\mathbb{R})$ is finer than the topology induced from $D_n^m(\mathbb{R})$.

Proof: We prove by induction over n. Consider the following cases:

1. $n = 0$. Then we have $D_0^{m+1}(\mathbb{R}) = D_0^n(\mathbb{R}) = \mathbb{R}$, and the statement holds;
2. $m = 0$. Then we have

$$D_n^1(\mathbb{R}) \subset \mathcal{C}(\mathbb{R}, D_{n-1}^0(\mathbb{R})) = \mathcal{C}(\mathbb{R}, C_{n-1}(\mathbb{R})) = C_n(\mathbb{R}) = D_n^0(\mathbb{R}),$$

   and the inclusion is continuous by the definition of differential topology.
3. $n, m > 0$, with the statement proven for $(m, n-1)$ and $(m-1, n)$. Let $f \in D_n^{m+1}(\mathbb{R})$. Then we have

$$f \in \mathcal{C}\big(\mathbb{R}, D_{n-1}^{m+1}(\mathbb{R})\big) \subset \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})).$$

   Moreover, since $f' \in D_n^m(\mathbb{R}) \subset D_n^{m-1}(\mathbb{R})$, we see that $f \in D_n^m(\mathbb{R})$ by definition, so we have the inclusion $D_n^{m+1}(\mathbb{R}) \subset D_n^m(\mathbb{R})$. It only remains to show that the inclusion map $j : D_n^{m+1}(\mathbb{R}) \to D_n^m(\mathbb{R})$ is continuous. This is equivalent to showing that the maps $i \circ j$ and $d \circ j$ are continuous, which is evident from the following commutative diagrams:

$$\begin{tikzcd} D_n^{m+1}(\mathbb{R}) \arrow[r, hook, "j"] \arrow[d, hook, "i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\ \mathcal{C}\big(\mathbb{R}, D_{n-1}^{m+1}(\mathbb{R})\big) \arrow[r, hook, "j"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \end{tikzcd} \qquad \begin{tikzcd} D_n^{m+1}(\mathbb{R}) \arrow[r, hook, "j"] \arrow[d, "d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\ D_n^m(\mathbb{R}) \arrow[r, hook, "j"'] & D_n^{m-1}(\mathbb{R}) \end{tikzcd}$$

■

Corollary 4.3. *We have a chain of subspaces with each topology finer than the previous one:*

$$C_n(\mathbb{R}) = D_n^0(\mathbb{R}) \overset{j}{\hookleftarrow} D_n^1(\mathbb{R}) \overset{j}{\hookleftarrow} D_n^2(\mathbb{R}) \overset{j}{\hookleftarrow} D_n^3(\mathbb{R}) \overset{j}{\hookleftarrow} \ldots$$

Lemma 4.4. *For all $n, m \in \mathbb{N}_0$, the topological space $D_n^m(\mathbb{R})$ is Hausdorff.*

Proof: Since $C_n(\mathbb{R})$ is Hausdorff and $i : D_n^m(\mathbb{R}) \to C_n(\mathbb{R})$ is continuous and injective, we see that $D_n(\mathbb{R})$ is also Hausdorff. ■

Lemma 4.5. *For all $m, n \in \mathbb{N}_0$, the pointwise linear structure is compatible with the topology on $D_n^m(\mathbb{R})$, making $D_n^m(\mathbb{R})$ a topological vector space.*

Proof: We prove by induction over n and m. If $n = 0$, we have $D_0(\mathbb{R}) = \mathbb{R}$ with the usual topology and linear structure. If $m = 0$, we have $D_n^0(\mathbb{R}) = C_n(\mathbb{R})$, which is a TVS. Suppose now that $m, n > 0$. We need to establish the continuity of the following two maps:

$$+ : D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \to D_n^m(\mathbb{R}), \qquad \cdot : \mathbb{R} \times D_n^m(\mathbb{R}) \to D_n^m(\mathbb{R}).$$

In turn, by the definition of the differential topology, this task is equivalent to proving that the maps $i \circ (+), d \circ (+), i \circ (\cdot), d \circ (\cdot)$ are continuous, where the maps i and d are defined in (2).
To this end, note that we have commutative diagrams

$$\begin{tikzcd} D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \arrow[r, "(+)"] \arrow[d, hook, "i \times i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\ \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \times \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \arrow[r, "(+)"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \end{tikzcd}$$

$$\begin{tikzcd}
D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \arrow[r, "(+)"] \arrow[d, "d \times d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\
D_n^{m-1}(\mathbb{R}) \times D_n^{m-1}(\mathbb{R}) \arrow[r, "(+)"'] & D_n^{m-1}(\mathbb{R})
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{R} \times D_n^m(\mathbb{R}) \arrow[r, "(\cdot)"] \arrow[d, hook, "\mathrm{id} \times i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\
\mathbb{R} \times \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \arrow[r, "(\cdot)"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{R} \times D_n^m(\mathbb{R}) \arrow[r, "(\cdot)"] \arrow[d, "\mathrm{id} \times d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\
\mathbb{R} \times D_n^{m-1}(\mathbb{R}) \arrow[r, "(\cdot)"'] & D_n^{m-1}(\mathbb{R})
\end{tikzcd}$$

Since the down-then-right path in each diagram is continuous, the right-then-down paths are continuous as well, and we are done. ■

Theorem 4.6. *Let $m, n \in \mathbb{N}$. A curried function $f \in H_n(\mathbb{R})$ lies in the class $D_n^m(\mathbb{R})$ if and only if its counterpart $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ has continuous partial derivatives on $\mathbb{R}^n$, of all orders up to m. Moreover, for each $1 \leq i_1 \leq i_2 \leq ... \leq i_m \leq n$, we have*

$$\begin{aligned} &\frac{\partial^m \tilde{f}}{\partial x_{i_1} \partial x_{i_2} ... \partial x_{i_m}}(x_1, x_2, ..., x_n) \\ &= f(x_1)...\left(x_{i_1-1}\right)'\left(x_{i_1}\right)...\left(x_{i_2-1}\right)'\left(x_{i_2}\right)...\left(x_{i_m-1}\right)'\left(x_{i_m}\right)...(x_n). \end{aligned} \tag{3}$$

Proof: If $n = 0$ or $m = 0$, the statement is trivial. Now let $n, m > 0$ and assume the statement proven for $(m, n-1)$ and $(m-1, n)$.

First, let $f \in D_n^m(\mathbb{R})$. We immediately see that $f \in D_n^{m-1}(\mathbb{R})$, and so, by the induction hypothesis, partial derivatives or order up to $m-1$ exist and are continuous on $\mathbb{R}^n$. Now, consider some $1 \leq i_1 \leq i_2 \leq ..., \leq i_m \leq n$. It is not hard to see by definition that (3) holds. To show that the partial derivative with respect to $x_{i_1}, ..., x_{i_m}$ is continuous, consider $(h_1, h_2, ..., h_n) \to 0 \in \mathbb{R}^n$, and a point $(x_1, ..., x_n) \in \mathbb{R}^n$. By the continuity of f and the definition of the differential topology, we have

$$\begin{aligned} f(x_1 + h_1) &\xrightarrow[h_1 \to 0]{} f(x_1) \in D_{n-1}^m(\mathbb{R}), \\ &\vdots \\ f(x_1 + h_1)...\left(x_{i_1-1} + h_{i_1-1}\right)' &\xrightarrow[h_1, ..., h_{i_1-1} \to 0]{} f(x_1)...\left(x_{i_1-1}\right)' \in D_{n-i_1+1}^{m-1}(\mathbb{R}), \\ &\vdots \\ &f(x_1 + h_1)...\left(x_{i_1-1} + h_{i_1-1}\right)'...\left(x_{i_m-1} + h_{i_m-1}\right)...(x_n + h_n) \\ &\xrightarrow[h_1, h_2, ..., h_n \to 0]{} f(x_1)...\left(x_{i_1-1}\right)'...\left(x_{i_m-1}\right)'...(x_n) \in D_0^0(\mathbb{R}) = \mathbb{R}, \end{aligned}$$

so the partial derivative

$$\frac{\partial^m \tilde{f}}{\partial x_{i_1} \partial x_{i_2} ... \partial x_{i_m}}$$

is continuous on $\mathbb{R}^n$.

Conversely, suppose $f \in H_n(\mathbb{R})$ is such that $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ lies in the class $C^m(\mathbb{R}^n)$. To show that $f \in D_n^m(\mathbb{R})$, we follow three steps:

1. For all $x_1 \in \mathbb{R}$, we have $f(x_1) \in D_{n-1}^m(\mathbb{R})$. This is provided by the induction hypothesis, since the restriction of a C^m function on a hyperplane is also a C^m function.
2. The function $f : \mathbb{R} \to D_{n-1}^m(\mathbb{R})$ is continuous. To show this, consider a sequence $\left\{x_k^{(1)}\right\}_{k \in \mathbb{N}} \subset \mathbb{R}$ converging to $x_0^{(1)} \in \mathbb{R}$. We aim to prove that $f\left(x_k^{(1)}\right) \to f\left(x_0^{(1)}\right)$ in $D_{n-1}^m(\mathbb{R})$. This is equivalent to showing that

$$f\left(x_k^{(1)}\right) \to f\left(x_k^{(1)}\right) \in \mathcal{C}(\mathbb{R}, D_{n-2}^m(\mathbb{R})) \quad \text{and} \quad f\left(x_k^{(1)}\right)' \to f\left(x_0^{(1)}\right)' \in D_{n-1}^{m-1}(\mathbb{R}). \tag{4}$$

   The latter of these conditions holds by the induction hypothesis, since

$$\frac{\partial \tilde{f}}{\partial x_2}(x_1, ..., x_n) = \frac{\partial \widetilde{f(x_1)}}{\partial x_2}(x_2, ..., x_n) \underset{\text{by (3)}}{=} f(x_1)'(x_2)...(x_n), \tag{5}$$

   and $\partial \tilde{f} / \partial x_2$ is continuous, which means that the map $x \mapsto f(x)'$ is continuous. To prove the former condition in (4), we again consider a sequence $x_k^{(2)} \to x_0^{(2)} \in \mathbb{R}$ and aim to prove

$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right) \in D_{n-2}(\mathbb{R}),$$

   which (unless $n = 2$) is again equivalent to

$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \to f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \quad \text{and} \quad f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right)' \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right)'.$$

   The latter follows from the induction hypothesis and the continuity of $\partial \tilde{f} / \partial x_3$, similarly to (5), and the former can again be expanded further. This chain leads us finally to the convergence

$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right)...\left(x_k^{(n)}\right) \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right)...\left(x_0^{(n)}\right) \in \mathbb{R},$$

   which is evident since $f \in C_n(\mathbb{R})$ by Proposition 3.5. Hence we see that f lies in $\mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R}))$. We have also established (3) for f and $i \geq 2$, via (5).
3. Finally, we need to find $f' \in D_n^{m-1}(\mathbb{R})$ such that

$$\frac{f(x+h) - f(x)}{h} \xrightarrow[h \to 0]{} f'(x) \in H_{n-1}(\mathbb{R}) \tag{6}$$

   for all $x \in \mathbb{R}$. Indeed, following (3), let us define $f' := \gamma\left(\partial \tilde{f} / \partial x_1\right)$. We immediately see by the induction hypothesis that f' so defined lies in $D_n^{m-1}(\mathbb{R})$. To see that (6) holds, consider $x_1, ..., x_n \in \mathbb{R}$ and write

$$\begin{aligned} f'(x_1)(x_2)...(x_n) &= \frac{\partial \tilde{f}}{\partial x_1}(x_1, ..., x_n) \\ &= \lim_{h \to 0} \frac{\tilde{f}(x_1 + h, x_2, ..., x_n) - \tilde{f}(x_1, ..., x_n)}{h} \\ &= \left(\lim_{h \to 0} \frac{f(x_1 + h) - f(x_1)}{h}\right)(x_2)...(x_n), \end{aligned}$$

   and we are done.

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A The language of filters

Definition A.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition A.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ *generated* by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.
If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x.
If $f : X \to Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X, then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition A.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \to x$, if it contains all open neighborhoods of x.

Remark A.4. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter
$$F = \{A \subset X \,|\, x_n \in A \text{ for all but finitely many } n\}$$
converges to x. This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.
- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \varnothing$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X.
- A function $f : X \to Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \to x \in X$ implies $f(F) \to f(x) \in Y$.

Lemma A.5. (see [1], page 74) *Let X be a set and $f_i : X \to Y_i$ be a family of maps, where Y_i are topological spaces. Let X be endowed with the coarsest topology such that f_i is continuous for each i. Then a filter $F \in \mathcal{F}(X)$ converges to $x \in X$ if and only if $f_i(F)$ converges to $f_i(x)$ for all i.*

B The topology of pointwise convergence

Definition B.1. (pointwise convergence topology, see [2]) Let X be a set, Y a topological space, and denote by $\mathcal{S}(X, Y)$ the set of functions from X to Y. For each $x \in X$ and each open set $U \subset Y$, let $V(x, U)$ be the set of all functions $f \in \mathcal{S}(X, Y)$ such that $f(x) \in U$. The collection
$$\{V(x, U) \,|\, x \in X, U \subset Y\}$$
forms a subbase of a topology on $\mathcal{S}(X, Y)$, called the *topology of pointwise convergence.*

Lemma B.2. *Let X be a set and Y a topological space. Then a filter $F \in \mathcal{F}(\mathcal{S}(X, Y))$ converges to a function $f \in \mathcal{S}(X, Y)$ if and only if, for any $x \in X$, the filter $F(x) = [\{A(x) \mid A \in F\}]$ converges to $f(x) \in Y$.*

Proof: First, let $F \to f \in \mathcal{S}(X,Y)$, and consider a point $x \in X$. To show that $F(x) \to f(x) \in Y$, let U be a neighborhood of $f(x)$. Then, since $F \to f$, we see that $V(x,U) \in F$. But this implies $U = V(x,U)(x) \in F(x)$, and so we conclude that $F(x) \to f(x)$.
Conversely, assume that $F(x) \to f(x)$ for all $x \in X$. To show that F converges to f, it suffices to show that $V(x,U) \in F$ whenever $f \in V(x,U)$. Indeed, if $f \in V(x,U)$, we have $f(x) \in U$, and so $U \in F(x)$, meaning that $U \supset A(x)$ for some $A \in F$, and so $V(x,U) \supset V(x,A(x)) \supset A \in F$, and we are done. ■

Lemma B.3. *Let X be a set and Y a Hausdorff topological space. Then $\mathcal{S}(X,Y)$ is also Hausdorff.*

Proof: Consider two functions $f, g \in \mathcal{S}(X,Y)$ such that $f \neq g$. This implies that there is $x_0 \in X$ such that $f(x_0) \neq g(x_0)$. Since Y is Hausdorff, there are neighborhoods $U_f \ni f(x_0)$ and $U_g \ni g(x_0)$ such that $U_f \cap U_g = \varnothing$. It then follows that $f \in V(x_0, U_f)$, $g \in V(x_0, U_g)$, and $V(x_0, U_f) \cap V(x_0, U_g) = \varnothing$. ■

Lemma B.4. *Let X be a set and let Y be a real topological vector space. Then $\mathcal{S}(X,Y)$ is also a real topological vector space with respect to the topology of pointwise convergence and the pointwise linear structure.*

Proof: We need to show that the maps $+ : \mathcal{S}(X,Y) \times \mathcal{S}(X,Y) \to \mathcal{S}(X,Y)$ and $\cdot : \mathbb{R} \times \mathcal{S}(X,Y) \to \mathcal{S}(X,Y)$ are continuous.
To begin, consider two filters $F_1, F_2 \in \mathcal{F}(\mathcal{S}(X,Y))$, converging to f_1 and f_2 respectively. To show that $F_1 + F_2$ converges to $f_1 + f_2$, consider $x \in X$. We have

$$\begin{aligned}(F_1 + F_2)(x) &= [\{(A+B)(x) \mid A \in F_1, B \in F_2\}] \\ &= [\{A(x) + B(x) \mid A \in F_1, B \in F_2\}] \\ &= F_1(x) + F_2(x) \to f_1(x) + f_2(x) = (f_1 + f_2)(x).\end{aligned}$$

This shows that the addition operation is continuous. Scalar multiplication is continuous by a similar argument. ■

C The compact-open topology

Definition C.1. (compact-open topology, see [2], page 301) Let X and Y be two topological spaces, and by $\mathcal{C}(X,Y)$ denote the set of all continuous functions from X to Y. For each compact set $K \subset X$ and each open set $U \subset Y$, let $V(K,U)$ be the set of all functions $f \in \mathcal{C}(X,Y)$ such that $f(K) \subset U$. The collection

$$\{V(K,U) \mid K \subset X, U \subset Y\}$$

forms a subbase of a topology on $\mathcal{C}(X,Y)$, called the *compact-open* topology.

Remark C.2. Clearly, the topology of pointwise convergence on $\mathcal{C}(X,Y)$ is coarser than the compact-open topology.

Proposition C.3. (see [3], page 34) *Let X and Y be topological spaces such that X is locally compact and Y is regular. Then a filter $F \in \mathcal{F}(\mathcal{C}(X,Y))$ converges to a function $f \in \mathcal{C}(X,Y)$ if and only if, for any filter $P \in \mathcal{F}(X)$ converging to $p \in X$, we have $F(P) \to f(p)$ in Y, where the filter $F(P)$ is based on*

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

Proposition C.4. (see [2], page 302) *Let X, Y, Z be topological spaces such that X is Hausdorff and Y is locally compact. Then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \to \mathcal{C}(X, \mathcal{C}(Y,Z))$$

defined as $\gamma(f)(x)(y) = f(x,y)$, is well-defined and is a homeomorphism.

Lemma C.5. *Let X, Y be topological spaces such that Y is Hausdorff. Then the space $\mathcal{C}(X,Y)$ is also Hausdorff.*

Proof: The compact-open topology on $\mathcal{C}(X,Y)$ is finer than the topology of pointwise convergence, and the latter is Hausdorff, which implies that the former is Hausdorff. ■

Lemma C.6. *Let X be a topological space and Y a real topological vector space. Then $\mathcal{C}(X,Y)$ is also a real topological vector space with respect to the pointwise linear structure and the compact-open topology.*

Proof: We will show that the map $+ : \mathcal{C}(X,Y) \times \mathcal{C}(X,Y) \to \mathcal{C}(X,Y)$ is continuous. Let $F_1, F_2 \in \mathcal{F}(\mathcal{C}(X,Y))$ converge to $f_1 \in \mathcal{C}(X,Y)$ and $f_2 \in \mathcal{C}(X,Y)$ respectively. For any filter $P \in \mathcal{F}(X)$ converging to $p \in X$, we have

$$\begin{aligned}(F_1+F_2)(P) &= [\{(A_1+A_2)(B) \mid A_i \in F_i, B \in P\}] \\ &\supset [\{A_1(B_1)+A_2(B_2) \mid A_i \in F_i, B_i \in P\}] \\ &= F_1(P)+F_2(P) \to f_1(p)+f_2(p) = (f_1+f_2)(p).\end{aligned}$$

The continuity of scalar multiplication is proven similarly. ■

Theorem C.7. (see [4], page 19) *A topological vector space X is metrizable if and only if it admits a countable local base of neighborhoods of 0.*

Lemma C.8. *Suppose that X is a locally compact and hemicompact topological space (that is, X is the union of countably many its compact subsets), and let Y be a Fréchet space. Then the space $\mathcal{C}(X,Y)$ is also a Fréchet space.*

Proof: First of all, we need to show that $\mathcal{C}(X,Y)$ is metrizable. Since Y is metrizable, by Theorem 3.7 it admits a countable basis $\{U_n\}_{n=1}^{\infty}$ of neighborhoods of 0. Moreover, since X is hemicompact, we have

$$X = \bigcup_{m=1}^{\infty} K_m,$$

where all K_m are compact. It is clear to see that the family

$$\{V(K_m, U_n) \mid m, n \in \mathbb{N}\}$$

forms a countable basis of neighborhoods of 0 in $\mathcal{C}(X,Y)$. Hence, by Theorem 3.7 the space $\mathcal{C}(X,Y)$ is metrizable.
It remains to show that the metric generating the topology of $\mathcal{C}(X,Y)$ is complete. To this end, consider a fundamental sequence $\{f_k\}_{k=1}^{\infty}$ in $\mathcal{C}(X,Y)$. This means that for all $n, m \in \mathbb{N}$, there is $N \in \mathbb{N}$ such that

$$k, l \geq N \Longrightarrow f_k - f_l \in V(K_m, U_n).$$

Now, consider a point $x \in X$. By choosing m_0 such that $x \in K_{m_0}$, we see that for all $n \in \mathbb{N}$, eventually

$$f_k - f_l \in V\Big(K_{m_0}, U_n\Big) \Longrightarrow f_k(x) - f_l(x) \in U_n.$$

This means that the sequence $\{f_k(x)\}_{k=1}^{\infty}$ is fundamental in Y, and so it converges to some point $f(x) \in Y$. We now need to show that f is continuous and $f_k \to f$ in $\mathcal{C}(X,Y)$.
Consider $n, m \in \mathbb{N}$. Since Y is a topological vector space, it is *regular* in the sense that every neighborhood U of 0 contains the closure $\overline{U'}$ of some other neighborhood U' of 0. Hence, we can find $n' \in \mathbb{N}$ such that $\overline{U_{n'}} \subset U_n$. Now, as $k, l \to \infty$, eventually we have

$$(f_k - f_l)(K_m) \subset U_{n'}.$$

Taking any $x \in K_m$, we have

$$f_k(x) - f_l(x) = (f_k - f_l)(x) \in U_{n'} \underset{l\to\infty}{\Longrightarrow} f_k(x) - f(x) \in \overline{U_{n'}} \subset U_n.$$

This means that

$$(f_k - f)(K_m) \subset U_n \tag{7}$$

eventually as $k \to \infty$. Now, to show that f is continuous, consider $x \in X$ and a neighborhood U of $0 \in Y$. Since X is locally compact, there is m_0 such that K_{m_0} is a neighborhood of x. Moreover, there is $n_0 \in \mathbb{N}$ such that $U_{n_0} + U_{n_0} + U_{n_0} \subset U$. Finally, by (7) there is $k_0 \in \mathbb{N}$ such that $\left(f_{k_0} - f\right)\left(K_{m_0}\right) \subset U_{n_0}$. Now, as $y \to x$, we have $y \in K_{m_0}$, and

$$f(y) - f(x) = [f(y) - f_k(y)] + [f_k(y) - f_k(x)] + [f_k(x) - f(x)] \in U_{n_0} + U_{n_0} + U_{n_0} \subset U.$$

Hence, we conclude that f is continuous, and (7) implies that for all $m, n \in \mathbb{N}$, we have $f_k - f \in V(K_m, U_n)$ eventually as $k \to \infty$, which means that $f_k \to f$ in $C(X, Y)$, as desired. ■